



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Department of Computer Science
Faculty of Computing

GROUP PROJECT

(Benchmarking of Computer Performance)

Programme : Bachelor of Computer Science
Subject Code : SECR1033
Subject Name : Computer Organization & Architecture
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Section : 02

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Video Link :

1. Demo on how benchmarking of CPU by using Freeware tools: <https://youtu.be/4HeL7BoBVIU>
2. Demo on running CPU benchmark using ASM code: https://youtu.be/t8kfUMkTe_E
3. Presentation of final project group (slide presentation): <https://youtu.be/fR-HERHLoXc>

Table of Contents

1.0 Introduction	3
2.0 Part 1	6
2.1 Flowchart of Project Part 1	6
2.2 Benchmarking Result	7
2.3 Benchmark Comparison	14
2.4 Discussion of Result	17
3.0 Part 2	19
3.1 Flowchart of Project Part 2	19
3.2 Coding and Implementation	20
3.3 Implementation Result	23
3.4 Analysis of Result	24
3.5 Discussion of Result	25
4.0 Conclusion	26
5.0 Reflection	27
6.0 References	28
7.0 Appendix	29

1.0 Introduction

Problem Background

The Central Processing Unit (CPU) is a key component of a computer that serves as its "control center". It interprets and executes instructions from the computer's memory, working with hardware components to complete tasks. Furthermore, the CPU is useful in determining the overall performance and efficiency of a system. Nowadays, new CPU models are always being released with improved architecture, therefore we must understand how to analyze and compare CPU performance in order to determine which brand of computer is the best purchase. CPU benchmarking is one of the tools that may be used in this evaluation process, allowing consumers to evaluate and understand the capabilities and limitations of various CPUs, as well as developers to design better CPUs.

Aim

The primary goal of this research is to provide a thorough introduction to CPU benchmarking, which provides insights into computer performance, hardware variances, and the complexities of software optimization. We will look at different benchmarks and tools for evaluating CPU performance. Furthermore, We'll also run a simple Assembly programme on three different computers and use functions to calculate the difference in CPU running time between the three.

Objective

1. **Understand the Principles of CPU Benchmarking** : Gain a foundational understanding of the concepts, importance and methodologies of CPU benchmarking.
2. **Compare the Performance of Different CPUs** : Evaluate and compare the performance of various CPUs to identify strengths and weaknesses.
3. **Analyse the Impact of Different Workloads on CPU Performance** : Assess how different types of workloads affect CPU performance.
4. **Learn to Use Free Benchmarking Tools and Interpret Their Results** : Acquire practical skills in using free benchmarking tools and learn how to interpret their outputs.

Scopes

Computer Specification

For computer to be benchmarked, we use Acer Aspire (A515-56), HP Laptop 15s-fq5xxx

Specifications	Computer Type		
	Acer Aspire 5	HP Laptop 15s-fq5xxx	ASUS Zenbook S13 OLED
Precessor Type	Intel Core i5 1135G7	Intel Core i7 1255U	AMD Ryzen 7 6800U
RAM Size	8 GBytes	32 GBytes	16 GBytes
GPU Type	Intel® Iris Xe Graphics	Intel® Iris Xe Graphics	AMD Radeon™ Graphics
Motherboard	TGL Iris TL (U3E1)	HP 8A20	AsusTEK COMPUTER INC. UM5302TA
Bus Specs.	PCI-Express 3.0 (8.0 GT/s)	PCI-Express 3.0 (8.0 GT/s)	PCI-Express 4.0 (16.0 GT/s)

Benchmarking Tools Features

1. CPU-Z

It is a lightweight tool that provides detailed information about the CPU, including its name, architecture, clock speeds, cache, and more. It also offers real-time monitoring of core frequencies and memory performance. The main uses of CPU-Z includes :

- Detailed System Specifications
- Real-Time Monitoring
- Diagnostics

2. Geekbench

It is a popular benchmarking tool that measures the single-core and multi-core performance of a CPU using a variety of simulated workloads. It provides a score that can be compared across different systems. The main uses of Geekbench includes :

- Standardized Performance Testing
- Comparative Analysis
- Performance Evaluation

3. Task Manager

The Task Manager in Windows provides real-time monitoring of CPU usage, clock speed, and thermal performance. It is a built-in utility that offers a quick overview of system resource utilization. The main uses of Task Manager includes :

- Performance Monitoring
- Process Management
- Troubleshooting

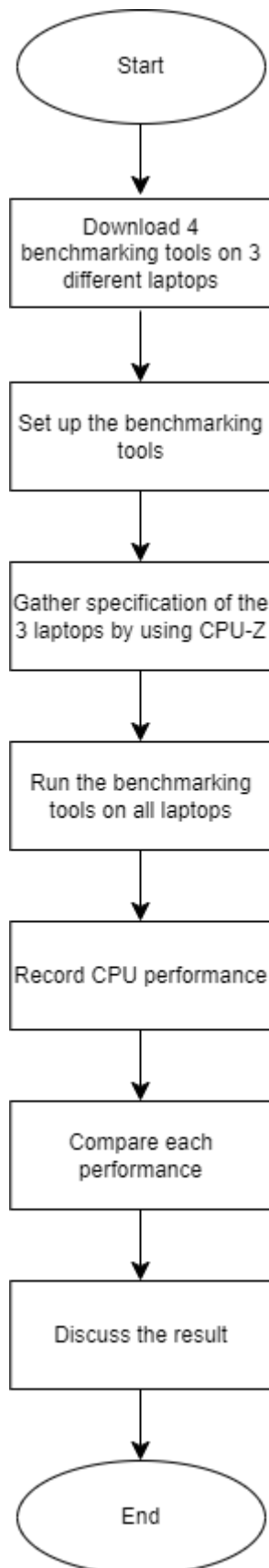
4. MiniTool Partition Wizard

While primarily a disk management tool, MiniTool Partition Wizard includes benchmarking features to test disk performance. It provides insights into read and write speeds, which can indirectly impact CPU performance. The main uses of MiniTool Partition Wizard includes :

- Storage Performance Testing
- Partition Management
- System Optimization

2.0 Part 1

2.1 Flowchart of Project Part 1



2.2 Benchmarking Result

CPU-Z

Benchmark	Computer Type		
	Acer Aspire 5	HP Laptop 15s-fq5xxx	ASUS Zenbook S13 OLED
CPU Technology	10nm	10nm	6nm
Number of Core	4	2P+8E	8
Core Speed	1305.12 MHz	4090.00 MHz	1393.61 MHz
Bus Speed	100.39 MHz	99.76 MHz	99.49 MHz
RAM Size	8 GBytes	32 GBytes	16 GBytes
GPU Size	-	-	-

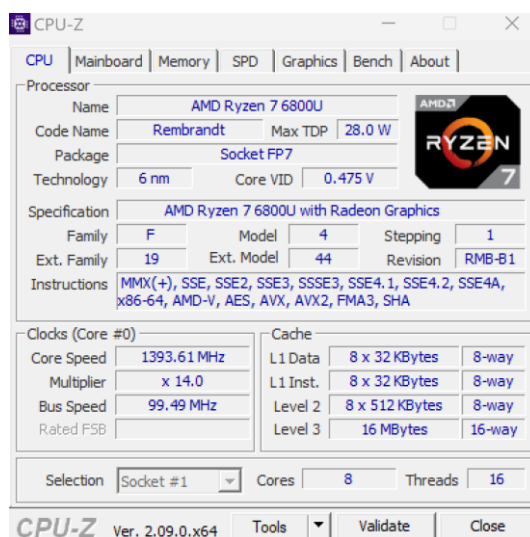
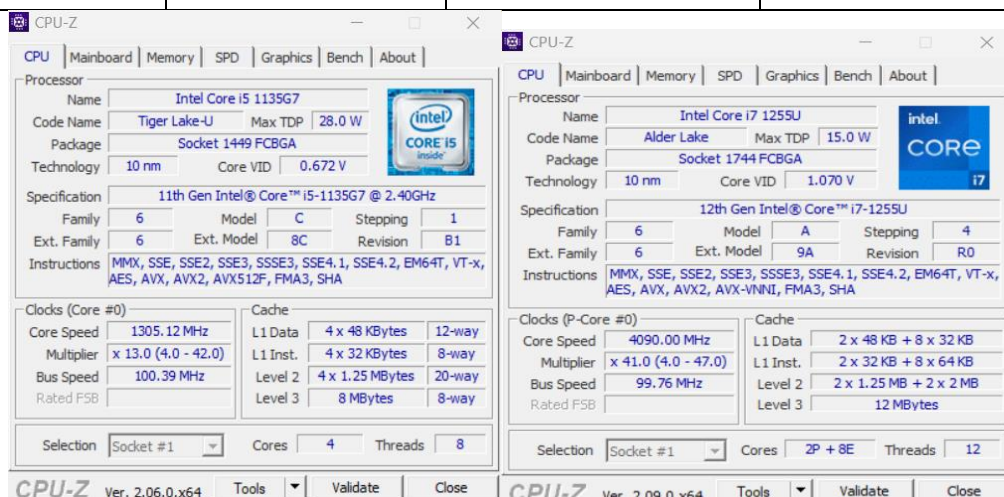


Figure A1: CPU-Z Output from 3 different computer

Geekbench

Acer Aspire A515-56	Geekbench 6 File Home CPU GPU Purchase Unlock Help News Benchmark Upgrade Help System Information Operating System: Microsoft Windows 11 Home Single Language (64-bit) Model: Acer Aspire A515-56 Motherboard: TGL Iris_TL Memory: 7.79 GB DDR4 SDRAM CPU Information Name: Intel Core i5-1135G7 Topology: 1 Processor, 4 Cores, 8 Threads Codename: Tiger Lake-U Package: Socket 1449 FCBGA Base Frequency: 2.40 GHz Maximum Frequency: 4201 MHz CPU Benchmark Geekbench measures CPU performance for everyday tasks using tests that model real-world applications. The CPU Benchmark takes several minutes to complete. Run CPU Benchmark CPU Information <table><tr><td>Name</td><td>Intel Core i5-1135G7</td></tr><tr><td>Topology</td><td>1 Processor, 4 Cores, 8 Threads</td></tr><tr><td>Identifier</td><td>GenuineIntel Family 6 Model 140 Stepping 1</td></tr><tr><td>Base Frequency</td><td>2.40 GHz</td></tr><tr><td>Cluster 1</td><td>4 Cores</td></tr><tr><td>Maximum Frequency</td><td>4201 MHz</td></tr><tr><td>Package</td><td>Socket 1449 FCBGA</td></tr><tr><td>Codename</td><td>Tiger Lake-U</td></tr><tr><td>L1 Instruction Cache</td><td>32.0 KB x 4</td></tr><tr><td>L1 Data Cache</td><td>48.0 KB x 4</td></tr><tr><td>L2 Cache</td><td>1.25 MB x 4</td></tr><tr><td>L3 Cache</td><td>8.00 MB x 1</td></tr></table>	Name	Intel Core i5-1135G7	Topology	1 Processor, 4 Cores, 8 Threads	Identifier	GenuineIntel Family 6 Model 140 Stepping 1	Base Frequency	2.40 GHz	Cluster 1	4 Cores	Maximum Frequency	4201 MHz	Package	Socket 1449 FCBGA	Codename	Tiger Lake-U	L1 Instruction Cache	32.0 KB x 4	L1 Data Cache	48.0 KB x 4	L2 Cache	1.25 MB x 4	L3 Cache	8.00 MB x 1		
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HP Laptop 15s-fq5xxx	Geekbench 6 File Home CPU GPU Purchase Unlock Help News Benchmark Upgrade Help System Information Operating System: Microsoft Windows 11 Home Single Language (64-bit) Model: HP HP Laptop 15s-fq5xxx Motherboard: HP 8A20 Memory: 32.0 GB DDR4 SDRAM CPU Information Name: Intel Core i7-1255U Topology: 1 Processor, 10 Cores, 12 Threads Codename: Alder Lake Package: Socket 1744 FCBGA Base Frequency: 2.59 GHz Maximum Frequency: 4688 MHz CPU Benchmark Geekbench measures CPU performance for everyday tasks using tests that model real-world applications. The CPU Benchmark takes several minutes to complete. Run CPU Benchmark CPU Information <table><tr><td>Name</td><td>Intel Core i7-1255U</td></tr><tr><td>Topology</td><td>1 Processor, 10 Cores, 12 Threads</td></tr><tr><td>Identifier</td><td>GenuineIntel Family 6 Model 154 Stepping 4</td></tr><tr><td>Base Frequency</td><td>2.59 GHz</td></tr><tr><td>Cluster 1</td><td>2 Cores</td></tr><tr><td>Cluster 2</td><td>8 Cores</td></tr><tr><td>Maximum Frequency</td><td>4688 MHz</td></tr><tr><td>Package</td><td>Socket 1744 FCBGA</td></tr><tr><td>Codename</td><td>Alder Lake</td></tr><tr><td>L1 Instruction Cache</td><td>32.0 KB x 6</td></tr><tr><td>L1 Data Cache</td><td>48.0 KB x 6</td></tr><tr><td>L2 Cache</td><td>1.25 MB x 1</td></tr><tr><td>L3 Cache</td><td>12.0 MB x 1</td></tr></table>	Name	Intel Core i7-1255U	Topology	1 Processor, 10 Cores, 12 Threads	Identifier	GenuineIntel Family 6 Model 154 Stepping 4	Base Frequency	2.59 GHz	Cluster 1	2 Cores	Cluster 2	8 Cores	Maximum Frequency	4688 MHz	Package	Socket 1744 FCBGA	Codename	Alder Lake	L1 Instruction Cache	32.0 KB x 6	L1 Data Cache	48.0 KB x 6	L2 Cache	1.25 MB x 1	L3 Cache	12.0 MB x 1
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ASUS Zenbook S13 OLED	Geekbench 6 File Home CPU GPU Purchase Unlock Help News Benchmark Upgrade Help System Information Operating System: Microsoft Windows 11 Home Single Language Model: ASUSTeK COMPUTER INC. Zenbook Motherboard: ASUSTeK COMPUTER INC. UM5302TA Memory: 15.2 GB DDR5 SDRAM CPU Information Name: AMD Ryzen 7 6800U Topology: 1 Processor, 8 Cores, 16 Threads Codename: Rembrandt Package: Socket FP7 Base Frequency: 2.70 GHz Maximum Frequency: 2470 MHz CPU Benchmark Geekbench measures CPU performance for everyday tasks using tests that model real-world applications. The CPU Benchmark takes several minutes to complete. Run CPU Benchmark CPU Information <table><tr><td>Name</td><td>AMD Ryzen 7 6800U</td></tr><tr><td>Topology</td><td>1 Processor, 8 Cores, 16 Threads</td></tr><tr><td>Identifier</td><td>AuthenticAMD Family 25 Model 68 Stepping 1</td></tr><tr><td>Base Frequency</td><td>2.70 GHz</td></tr><tr><td>Cluster 1</td><td>8 Cores</td></tr><tr><td>Maximum Frequency</td><td>2469 MHz</td></tr><tr><td>Package</td><td>Socket FP7</td></tr><tr><td>Codename</td><td>Rembrandt</td></tr><tr><td>L1 Instruction Cache</td><td>32.0 KB x 8</td></tr><tr><td>L1 Data Cache</td><td>32.0 KB x 8</td></tr><tr><td>L2 Cache</td><td>512 KB x 8</td></tr><tr><td>L3 Cache</td><td>16.0 MB x 1</td></tr></table>	Name	AMD Ryzen 7 6800U	Topology	1 Processor, 8 Cores, 16 Threads	Identifier	AuthenticAMD Family 25 Model 68 Stepping 1	Base Frequency	2.70 GHz	Cluster 1	8 Cores	Maximum Frequency	2469 MHz	Package	Socket FP7	Codename	Rembrandt	L1 Instruction Cache	32.0 KB x 8	L1 Data Cache	32.0 KB x 8	L2 Cache	512 KB x 8	L3 Cache	16.0 MB x 1		
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Table 2: Greekbench Output of 3 different computer

Greekbench -CPU Load Stress Test (CPU Test)

Acer Aspire A515-56	Single-Core Performance		Multi-Core Performance	
	Single-Core Score	1163	Multi-Core Score	3346
	File Compression	1020 146.5 MB/sec	File Compression	1435 206.1 MB/sec
	Navigation	1284 7.74 routes/sec	Navigation	4714 28.4 routes/sec
	HTML5 Browser	1060 21.7 pages/sec	HTML5 Browser	1855 38.0 pages/sec
	PDF Renderer	990 22.8 Mpxels/sec	PDF Renderer	3524 81.3 Mpxels/sec
	Photo Library	1169 15.9 images/sec	Photo Library	4782 64.9 images/sec
	Clang	1041 5.13 Klines/sec	Clang	5753 28.3 Klines/sec
	Text Processing	1118 89.5 pages/sec	Text Processing	1491 119.4 pages/sec
	Asset Compression	1321 40.9 MB/sec	Asset Compression	6585 204.0 MB/sec
	Object Detection	1329 39.8 images/sec	Object Detection	1840 55.0 images/sec
	Background Blur	1752 7.25 images/sec	Background Blur	5037 20.8 images/sec
	Horizon Detection	1413 44.0 Mpxels/sec	Horizon Detection	4759 148.1 Mpxels/sec
	Object Remover	838 64.4 Mpxels/sec	Object Remover	3315 254.9 Mpxels/sec
	HDR	1313 38.5 Mpxels/sec	HDR	3712 108.9 Mpxels/sec
	Photo Filter	1182 11.7 images/sec	Photo Filter	2622 26.0 images/sec
	Ray Tracer	888 858.9 Kpxels/sec	Ray Tracer	5317 5.14 Mpxels/sec
	Structure from Motion	1301 41.2 Kpxels/sec	Structure from Motion	4029 127.6 Kpxels/sec

HP Laptop 15s-fq5xxx	Single-Core Performance			
	Single-Core Score	2029		
	File Compression	1604 230.3 MB/sec		
	Navigation	1620 9.76 routes/sec		
	HTML5 Browser	1893 38.8 pages/sec		
	PDF Renderer	1826 42.1 Mpixels/sec		
	Photo Library	1800 24.4 images/sec		
	Clang	2144 10.6 Klines/sec		
	Text Processing	2141 171.4 pages/sec		
	Asset Compression	2072 64.2 MB/sec		
	Object Detection	2184 65.4 images/sec		
	Background Blur	2902 12.0 images/sec		
	Horizon Detection	2794 86.9 Mpixels/sec		
	Object Remover	1926 148.1 Mpixels/sec		
	HDR	2278 66.9 Mpixels/sec		
	Photo Filter	2504 24.8 images/sec		
	Ray Tracer	1877 1.82 Mpixels/sec		
	Structure from Motion	1890 59.9 Kpixels/sec		
	Multi-Core Performance	Multi-Core Score		6005
		File Compression	3851 553.1 MB/sec	
Navigation		7982 48.1 routes/sec		
HTML5 Browser		7365 150.8 pages/sec		
PDF Renderer		7623 175.8 Mpixels/sec		
Photo Library		6138 83.3 images/sec		
Clang		6046 29.8 Klines/sec		
Text Processing		2152 172.3 pages/sec		
Asset Compression		8111 251.3 MB/sec		
Object Detection		3785 113.3 images/sec		
Background Blur		6796 28.1 images/sec		
Horizon Detection		8236 256.3 Mpixels/sec		
Object Remover		5873 451.6 Mpixels/sec		
HDR		6183 181.4 Mpixels/sec		
Photo Filter		6385 63.4 images/sec		
Ray Tracer		9599 9.29 Mpixels/sec		
Structure from Motion		8002 253.4 Kpixels/sec		

ASUS Zenbook S13 OLED	Single-Core Performance	
	Single-Core Score	1102
	File Compression	1195 171.0 MB/sec
	Navigation	1059 6.38 routes/sec
	HTML5 Browser	1111 22.7 pages/sec
	PDF Renderer	1079 24.9 Mpixels/sec
	Photo Library	1000 13.6 images/sec
	Clang	1065 5.25 Klines/sec
	Text Processing	1055 84.5 pages/sec
	Asset Compression	1119 34.7 MB/sec
	Object Detection	586 17.5 images/sec
	Background Blur	1545 6.39 images/sec
	Horizon Detection	1484 46.2 Mpixels/sec
	Object Remover	1110 85.4 Mpixels/sec
	HDR	1296 38.0 Mpixels/sec
	Photo Filter	1478 14.7 images/sec
	Ray Tracer	1002 969.4 Kpixels/sec
	Structure from Motion	1220 38.6 Kpixels/sec

Multi-Core Performance	Multi-Core Score		6125
	File Compression	4478 643.0 MB/sec	
	Navigation	7931 47.8 routes/sec	
	HTML5 Browser	6788 139.0 pages/sec	
	PDF Renderer	7722 178.1 Mpixels/sec	
	Photo Library	6082 82.5 images/sec	
	Clang	9091 44.8 Klines/sec	
	Text Processing	1363 109.2 pages/sec	
	Asset Compression	10096 312.6 MB/sec	
	Object Detection	2792 83.5 images/sec	
	Background Blur	7724 32.0 images/sec	
	Horizon Detection	8975 279.3 Mpixels/sec	
	Object Remover	5548 426.5 Mpixels/sec	
	HDR	7776 228.2 Mpixels/sec	
	Photo Filter	5917 58.7 images/sec	
	Ray Tracer	11648 11.3 Mpixels/sec	
	Structure from Motion	7910 250.4 Kpixels/sec	

Table 3: Greekbench CPU Load Stress Test (CPU Test) of 3 different computer

Window Task Manager

Computer Type	Workloads																																																																																																																																																																																																		
	High CPU Load	Mixed Load	Low CPU Load																																																																																																																																																																																																
Acer Aspire A515-56	Processes Run new task End task Efficiency mode <table><tr><th>Name</th><th>Status</th><th>CPU</th><th>Memory</th><th>Disk</th><th>Network</th></tr><tr><td colspan="6">Apps (3)</td></tr><tr><td>CapCut (1)</td><td></td><td>24.8%</td><td>3,563.5 MB</td><td>0.34 MB/s</td><td>0.1 Mbps</td></tr><tr><td>Microsoft Edge (17)</td><td></td><td>4%</td><td>386.6 MB</td><td>0 MB/s</td><td>0 Mbps</td></tr><tr><td>Task Manager</td><td></td><td>1.5%</td><td>76.8 MB</td><td>0 MB/s</td><td>0 Mbps</td></tr><tr><td colspan="6">Background processes (76)</td></tr><tr><td>ACCTM</td><td></td><td>4%</td><td>5.8 MB</td><td>0.1 MB/s</td><td>0 Mbps</td></tr><tr><td>ACCSvc</td><td></td><td>4%</td><td>0.3 MB</td><td>0 MB/s</td><td>0 Mbps</td></tr><tr><td>Acrobat Update Service (32 bit)</td><td></td><td>4%</td><td>0.6 MB</td><td>0 MB/s</td><td>0 Mbps</td></tr><tr><td>Agent for EasyConnect (32 bit)</td><td></td><td>4%</td><td>2.9 MB</td><td>0 MB/s</td><td>0 Mbps</td></tr></table>	Name	Status	CPU	Memory	Disk	Network	Apps (3)						CapCut (1)		24.8%	3,563.5 MB	0.34 MB/s	0.1 Mbps	Microsoft Edge (17)		4%	386.6 MB	0 MB/s	0 Mbps	Task Manager		1.5%	76.8 MB	0 MB/s	0 Mbps	Background processes (76)						ACCTM		4%	5.8 MB	0.1 MB/s	0 Mbps	ACCSvc		4%	0.3 MB	0 MB/s	0 Mbps	Acrobat Update Service (32 bit)		4%	0.6 MB	0 MB/s	0 Mbps	Agent for EasyConnect (32 bit)		4%	2.9 MB	0 MB/s	0 Mbps	Processes Run new task End task Efficiency mode <table><tr><th>Name</th><th>Status</th><th>CPU</th><th>Memory</th><th>Disk</th><th>Network</th></tr><tr><td colspan="6">Apps (4)</td></tr><tr><td>Google Chrome (32 bit) (10)</td><td></td><td>2.4%</td><td>535.4 MB</td><td>0.1 MB/s</td><td>0.1 Mbps</td></tr><tr><td>Microsoft Edge (16)</td><td></td><td>0%</td><td>278.1 MB</td><td>0 MB/s</td><td>0 Mbps</td></tr><tr><td>Microsoft Word</td><td></td><td>0%</td><td>153.7 MB</td><td>0 MB/s</td><td>0 Mbps</td></tr><tr><td>Task Manager</td><td></td><td>1.1%</td><td>69.6 MB</td><td>0.1 MB/s</td><td>0 Mbps</td></tr><tr><td colspan="6">Background processes (78)</td></tr><tr><td>ACCTM</td><td></td><td>0%</td><td>2.7 MB</td><td>0 MB/s</td><td>0 Mbps</td></tr><tr><td>ACCSvc</td><td></td><td>0%</td><td>0.3 MB</td><td>0 MB/s</td><td>0 Mbps</td></tr><tr><td>Acrobat Update Service (32 bit)</td><td></td><td>0%</td><td>0.6 MB</td><td>0 MB/s</td><td>0 Mbps</td></tr><tr><td>Agent for EasyConnect (32 bit)</td><td></td><td>0%</td><td>2.9 MB</td><td>0 MB/s</td><td>0 Mbps</td></tr></table>	Name	Status	CPU	Memory	Disk	Network	Apps (4)						Google Chrome (32 bit) (10)		2.4%	535.4 MB	0.1 MB/s	0.1 Mbps	Microsoft Edge (16)		0%	278.1 MB	0 MB/s	0 Mbps	Microsoft Word		0%	153.7 MB	0 MB/s	0 Mbps	Task Manager		1.1%	69.6 MB	0.1 MB/s	0 Mbps	Background processes (78)						ACCTM		0%	2.7 MB	0 MB/s	0 Mbps	ACCSvc		0%	0.3 MB	0 MB/s	0 Mbps	Acrobat Update Service (32 bit)		0%	0.6 MB	0 MB/s	0 Mbps	Agent for EasyConnect (32 bit)		0%	2.9 MB	0 MB/s	0 Mbps	Processes Run new task End task Efficiency mode <table><tr><th>Name</th><th>Status</th><th>CPU</th><th>Memory</th><th>Disk</th><th>Network</th></tr><tr><td colspan="6">Apps (4)</td></tr><tr><td>Microsoft Edge (17)</td><td></td><td>0%</td><td>339.5 MB</td><td>0 MB/s</td><td>0 Mbps</td></tr><tr><td>Microsoft Word</td><td></td><td>0.2%</td><td>76.4 MB</td><td>0 MB/s</td><td>0 Mbps</td></tr><tr><td>Photos</td><td></td><td>0.4%</td><td>254.9 MB</td><td>0 MB/s</td><td>0 Mbps</td></tr><tr><td>Task Manager</td><td></td><td>0.7%</td><td>71.3 MB</td><td>0 MB/s</td><td>0 Mbps</td></tr><tr><td colspan="6">Background processes (77)</td></tr><tr><td>ACCTM</td><td></td><td>0%</td><td>3.0 MB</td><td>0 MB/s</td><td>0 Mbps</td></tr><tr><td>ACCSvc</td><td></td><td>0%</td><td>0.3 MB</td><td>0 MB/s</td><td>0 Mbps</td></tr><tr><td>Acrobat Update Service (32 bit)</td><td></td><td>0%</td><td>0.6 MB</td><td>0 MB/s</td><td>0 Mbps</td></tr><tr><td>Agent for EasyConnect (32 bit)</td><td></td><td>0%</td><td>2.9 MB</td><td>0 MB/s</td><td>0 Mbps</td></tr></table>	Name	Status	CPU	Memory	Disk	Network	Apps (4)						Microsoft Edge (17)		0%	339.5 MB	0 MB/s	0 Mbps	Microsoft Word		0.2%	76.4 MB	0 MB/s	0 Mbps	Photos		0.4%	254.9 MB	0 MB/s	0 Mbps	Task Manager		0.7%	71.3 MB	0 MB/s	0 Mbps	Background processes (77)						ACCTM		0%	3.0 MB	0 MB/s	0 Mbps	ACCSvc		0%	0.3 MB	0 MB/s	0 Mbps	Acrobat Update Service (32 bit)		0%	0.6 MB	0 MB/s	0 Mbps	Agent for EasyConnect (32 bit)		0%	2.9 MB	0 MB/s	0 Mbps
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Summarize	High CPU workloads, like video rendering, use a lot of processing power, making the CPU work hard and get hot. Memory and disk use are high, and network use can be heavy if transferring large data.	Mixed workloads, like compiling code while running other applications, have varying CPU demands, causing fluctuating CPU use. Memory and disk use are moderate with some spikes. Network use varies, balanced but can be heavy at times.	Low CPU workloads, like basic office tasks, use little processing power, keeping the CPU cool and responsive. Memory and disk use are low, and network use is minimal, making the system quick and efficient.
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Table 4: Process versus resources usage (CPU,Memory,Disk,Network,GPU) of 3 different computer

Computer Processor Types	ModeTest SequentialRandom	
	Acer Aspire A515-56	
HP Laptop 15s-fq5xxx		
ASUS Zenbook S13 OLED		

2.3 Benchmark Comparison

Computer	CPU-Z (System Info)	Geekbench (CPU stress test)	Task Manager (Active Program and Resource Adjustment)	MiniTools (Disk benchmark)
Acer Aspire A515-56	CPU Technology : 10nm Number of Cores: 4 Core Speed : 1305.12 MHz Bus Speed : 100.39 MHz RAM Size : 8 GBytes	Single - Core : 1163 Multi - Core : 3346	High CPU Load: CPU: 31% Memory: 84% Disk: 21% Network: 0% Mixed Load: CPU: 7% Memory: 75% Disk: 0% Network: 0% Low CPU Load: CPU: 2% Memory: 73% Disk: 0% Network: 0%	Sequential Read Performance: Acer Aspire A515-56 shows the lowest sequential read speeds across all block sizes, indicating it might be less efficient for read-intensive tasks. Sequential Write Performance: Acer Aspire A515-56 demonstrates strong sequential write performance at higher block sizes but falls behind HP for smaller sizes. Random Read Performance: Acer Aspire A515-56 lags in random read speeds compared to ASUS and HP, which may impact its efficiency in scenarios requiring random access. Random Write Performance: Acer Aspire A515-56 performs well at higher block sizes but falls short at smaller ones.
HP Laptop 15s-fq5xxx	CPU Technology: 10nm Number of Cores: 2P+8E Core Speed: 4090.00 MHz	Single - Core: 2029 Multi - Core: 6005	High CPU Load: CPU: 13% Memory: 32% Disk: 3% Network: 0% Mixed Load: CPU: 12% Memory: 32%	Sequential Read Performance: HP Laptop 15s-fq5xxx performs well across all block sizes, maintaining high read speeds, particularly at 32 KB (1892.66 MB/s) and 16 KB (1913.96 MB/s).

	Bus Speed: 99.76 MHz RAM Size: 32 GBytes		Disk: 4% Network: 0% Low CPU Load: CPU: 2% Memory: 30% Disk: 5% Network: 0%	Sequential Write Performance: HP Laptop 15s-fq5xxx leads with the highest sequential write speeds at 32 KB (2527.83 MB/s) and performs consistently well across different block sizes. Random Read Performance: HP Laptop 15s-fq5xxx also delivers strong random read performance, especially at higher block sizes (32 KB: 1819.16 MB/s). Random Write Performance: HP Laptop 15s-fq5xxx leads again in random write performance at 32 KB (2423.92 MB/s) and maintains good speeds across other block sizes.
ASUS Zenbook S13 OLED	CPU Technology : 6nm Number of Cores: 8 Core Speed : 1393.61 MHz Bus Speed : 99.49 MHz RAM Size : 16 GBytes	Single - Core : 1102 Multi - Core : 6125	High CPU Load: CPU: 5% Memory: 47% Disk: 2% Network: 1% Mixed Load: CPU: 7% Memory: 46% Disk: 3% Network: 1% Low CPU Load: CPU: 2% Memory: 42% Disk: 1% Network: 1%	Sequential Read Performance: ASUS Zenbook S13 OLED exhibits the highest sequential read speeds at 32 KB (2548.21 MB/s) and 4 KB (611.317 MB/s), though it drops for intermediate block sizes. Sequential Write Performance: ASUS Zenbook S13 OLED has the lowest sequential write performance, particularly evident at 4 KB (250.595 MB/s), making it less ideal for write-heavy operations. Random Read Performance: ASUS Zenbook S13 OLED shines with the highest random read speeds, notably at 32 KB

				(2583.67 MB/s) and 4 KB (610.731 MB/s). Random Write Performance: ASUS Zenbook S13 OLED shows lower random write speeds, especially noticeable at 4 KB (248.885 MB/s).
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2.4 Discussion of Result

CPU-Z (System Info)

Acer Aspire 5, with its Intel Core i5 processor, is built on a 10 nm process technology and feature 4 cores. This setup is adequate for basic to moderate computing tasks, such as web browsing, office applications and light multitasking. The 8 GB of RAM supports these activities, but the lower core count and core speed mean it may struggle with more intensive applications or multitasking scenarios. This suitable for general productivity and everyday use but not ideal for demanding workloads. **HP Laptop 15s-fq5xxx**, features a more advanced Intel Core i7 processor with a unique combination of 2 performance cores and 8 efficiency cores. The high core speed (4090.00 MHz) and large RAM (32 GB) make it well-suited for demanding tasks, such as software development, data analysis and multimedia editing. The combination of performance and efficiency cores also provides a balance between power consumption and performance. This is high efficiency for multitasking and resource-intensive applications, making it a versatile choice for professionals. **ASUS Zenbook S13 OLED**, powered by an AMD Ryzen 7 processor, uses a 6nm process technology with 8 cores. This configuration provides excellent performance for both single-threaded and multi-threaded applications. The 16 GB of RAM is sufficient for most professional tasks and heavy multitasking. The advanced 6nm technology suggests better energy efficiency and performance density compared to the 10nm process in the other laptops. This offers the best overall performance and efficiency among the three, making it the **ideal choice for power users and professionals**.

Geekbench (CPU Stress Test)

Acer Aspire 5 shows lower single-core and multi-core scores among the three laptops. This indicates that it is sufficient for basic computing tasks but may not perform well in more demanding scenarios that require significant processing power or multitasking capabilities. The lower cores are reflective of its fewer cores and lower overall performance. **HP Laptop 15s-fq5xxx** achieve the highest single-core score and a solid multi-core score. This suggest strong performance in tasks that require fast, single-threaded processing, as well as good capability for handling multitasking and parallel processing efficiently. This makes it a strong contender for users who required both high single-thread and multi-thread performance. **ASUS Zenbook S13 OLED** has the highest multi-core score, indicating its superior performance in multi-threaded tasks. While its single-core score is slightly lower than the Acer Aspire 5, its multi-core performance makes it ideal for applications that can leverage multiple cores, such as video rendering and complex simulations.

Task Manager (Active Program and Resource Adjustment)

ASUS ZenBook S13 OLED offers the best overall performance, thanks to its advanced 6nm CPU technology which appear to be more efficient than a 10nm process for CPU technology because smaller process sizes pack more transistors into the chip, resulting in better performance and lower power consumption. Other than that ,8 cores and 16GB of RAM, providing excellent CPU efficiency and moderate memory usage across all workloads. **HP Laptop 15s-fq5xxx** also performs well, especially with its substantial 32GB RAM, making it highly efficient in memory management but it has higher CPU usage compared to ASUS under high load. **Acer Aspire 5**, while capable, shows higher CPU and memory usage due to its 10nm 4-core CPU and 8GB RAM, making it less efficient under heavy loads compared to the other two laptops. Thus, **ASUS Zenbook S13 OLED** as leaders in performance and efficiency among these models.

MiniTools (Disk Benchmark)

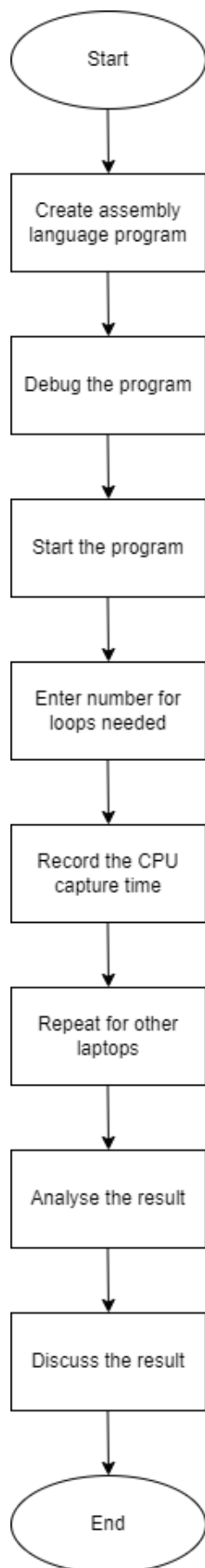
HP Laptop 15s-fq5xxx has the best overall disk performance as it offers superior write performance and maintains strong read speeds, making it versatile for both read and write-intensive applications.

ASUS Zenbook S13 OLED excels in read operations, both sequential and random, making it suitable for tasks demanding high read speeds but lags in write performance, especially at smaller block sizes.

Acer Aspire 5, while not leading in any category, offers a balanced performance suitable for general usage but may not be ideal for high-performance tasks.

3.0 Part 2

3.1 Flowchart of Project Part 2



3.2 Coding and Implementation

TITLE COA Project

; Author:
; 1. Evelyn Goh Yuan Qi
; 2. Joanne Ching Yin Xuan
; 3. Lim Yu Han

; Date: 03 July 2024

include Irvine32.inc

.data

msgWelcome **BYTE** "Welcome to CPU Benchmark Program", 0dh, 0ah, 0
msgAlgorithm **BYTE** "Benchmark CPU time Using Equation $y = (11*x^3) + (17*x^2) + (13*x) + 20$ ", 0ah, 0dh
BYTE "(with delay coef1, coef2, coef3, coef4 = 11, 17, 13, 20 msec)", 0dh, 0ah, 0
msgEnterN **BYTE** "Enter Number of Looping (N) = ", 0
msgProgress **BYTE** "CPU time Stress Test in progress...", 0dh, 0ah, 0
msgResult **BYTE** "Result: ", 0dh, 0ah, 0
msgFirstCapture **BYTE** "First Capture Execution time in millisecond: ", 0
msgSecondCapture **BYTE** "Second Capture Execution time in millisecond: ", 0
msgDiffTime **BYTE** "Different Execution time in millisecond: ", 0
msgSum **BYTE** "Value of Sum from the Stress Test (polynomial) = ", 0
msgPrompt **BYTE** "Press 'y' to continue or 'n' to exit the benchmark: ", 0
msgBye **BYTE** "Thank you ... BYE!!", 0dh, 0ah, 0

coef1 **DWORD** 11; a = 11
coef2 **DWORD** 17; b = 17
coef3 **DWORD** 13; c = 13
coef4 **DWORD** 20; d = 20

max_loop **DWORD** ?
sum **WORD** ?
capture_msec_before **DWORD** ?
capture_msec_after **DWORD** ?
elapsed_time **DWORD** ?

.code

main **PROC**
; Display welcome message
call clrscr
mov edx, **offset** msgWelcome
call WriteString
call crlf

; Display the algorithm
mov edx, **offset** msgAlgorithm
call WriteString
call crlf

; Prompt for max loop value
mov edx, **offset** msgEnterN
call WriteString
call ReadDec
mov max_loop, eax

; Display progress message
mov edx, **offset** msgProgress
call WriteString
call crlf

; Capture time before starting the loop
call GetMseconds
mov capture_msec_before, eax

```

; Initialize sum to 0
mov sum, 0

; Set loop counter
mov ecx, max_loop
mov ebx, 1

calc_loop:
; Calculate y = (coef1 * x ^ 3) + (coef2 * x ^ 2) + (coef3 * x) + coef4
; coef1* x ^ 3
mov eax, coef1
call Delay
mul bx
mul bx
mul bx
mov dx, ax; Store in dx for adding later

; coef2* x ^ 2
mov eax, coef2
call Delay
mul bx
mul bx
add dx, ax; Add previous result

; coef3* x
mov eax, coef3
call Delay
mul bx
add dx, ax; Add previous result

; coef4
mov eax, coef4
call Delay
add dx, ax; Add coef4

; Add y to sum
add sum, dx

; Increment loop counter
inc ebx
loop calc_loop

; Capture time after completing the loop
call GetMseconds
mov capture_msec_after, eax

; Calculate elapsed time
mov eax, capture_msec_after
sub eax, capture_msec_before
mov elapsed_time, eax

; Display results
mov edx, offset msgResult
call WriteString
call crlf

mov edx, offset msgFirstCapture
call WriteString
mov eax, capture_msec_before
call WriteDec
call crlf

mov edx, offset msgSecondCapture
call WriteString
mov eax, capture_msec_after
call WriteDec
call crlf

mov edx, offset msgDiffTime

```

```
call WriteString
mov eax, elapsed_time
call WriteDec
call crlf

mov edx, offset msgSum
call WriteString
mov ax, sum
call WriteDec
call crlf

; Prompt to continue or exit
mov edx, offset msgPrompt
call WriteString
call ReadChar

cmp al, 'y'
je main

mov edx, offset msgBye
call WriteString

; Exit program
exit
main ENDP
END main
```

3.3 Implementation Result

Computer Type	No of Loop (N)	Capture_msec (before)	Capture_msec (after)	CPU_msec (different)
Acer Aspire 5	50	999977	1004667	4690
	100	1028282	1037591	9309
	200	1063048	1081688	18640
	500	1113000	1159582	46582
	700	1181983	1247264	65281
	1000	1294862	1387984	93122
HP Laptop 15s-fq5xxx	50	43171501	43176224	4723
	100	43294987	43304329	9342
	200	43368293	43387096	18803
	500	43434571	43481681	47110
	700	43526607	43592072	65465
	1000	43641527	43734931	93404
ASUS Zenbook S13 OLED	50	64761981	64766709	4728
	100	64838823	64848233	9410
	200	64897692	64916448	18756
	500	64963217	65010237	47020
	700	65342729	65408408	65679
	1000	65479047	65572837	93790

3.4 Analysis of Result

- Acer Aspire 5

The CPU time (**CPU_msec**) increases as the number of loops (**No of Loop**) increases. The difference in capture time before and after (**Capture_msec (before)**) and (**Capture_msec (after)**) also increases with the number of loops. The trend shows that the processing time roughly scales with the number of loops, indicating that the CPU handles more iterations by taking proportionally more time.

- HP Laptop 15s-fq5xxx

Similar to Acer Aspire 5, the CPU time increases with the number of loops. The capture times before and after also increase proportionally to the number of loops. The differences in capture time before and after are relative similar to those observed in the Acer Aspire 5, suggesting a consistent increase in processing time with increased workload.

- Asus Zenbook S13 OLED

The trend of increasing CPU time with more loops continues. The capture time before and after execution show a consistent increase with the number of loops. The difference in capture time follow a pattern similar to the other two laptops, indicating a proportional relationship between the number of loops and the processing time.

3.5 Discussion of Result

The implementation results for the CPU testing program across three different laptops, including the Acer Aspire 5, HP Laptop 15s-fq5xxx, and ASUS Zenbook S13 OLED, provide a complete perspective of their performance under various workloads. The results reveal a clear pattern in which CPU time, represented by the difference in milliseconds, increases as the number of loops grows. This pattern is consistent across all three devices, demonstrating that each laptop's processing time varies proportionally with the workload.

For the Acer Aspire 5, the CPU time difference begins at 4690 milliseconds for 50 cycles and increases to 93122 milliseconds for 1000 loops. This huge rise indicates that, while the laptop can handle greater workloads, the overhead becomes increasingly noticeable as the number of repeats increases. Similarly, the CPU time differential for the HP Laptop 15s-fq5xxx increases from 4723 milliseconds for 50 cycles to 93404 milliseconds for 1,000 loops. This laptop's performance is equivalent to the Acer Aspire 5, with somewhat lower CPU time disparities at larger loop counts. The Asus Zenbook S13 OLED continues this trend, although with slightly lower CPU time disparities throughout all rounds, beginning at 4728 milliseconds for 50 cycles and increasing to 93790 milliseconds for 1000 loops. This suggests that this laptop may be more efficient at managing growing workloads, possibly due to improved hardware specs or increased system performance.

In a nutshell all three laptops show predictable scaling in processing time with higher workloads. However, the Asus Zenbook S13 OLED looks to handle the workload more efficiently, making it an excellent choice for computationally intensive activities.

4.0 Conclusion

In this project, we conducted a comprehensive benchmarking analysis of three different laptops: the Acer Aspire 5, HP Laptop 15s-fq5xxx, and ASUS Zenbook S13 OLED. By utilizing various benchmarking tools such as CPU-Z, Geekbench, Task Manager, and MiniTool Partition Wizard, we evaluated the performance of these laptops under different workloads. Our analysis revealed significant insights into their CPU efficiency, memory management, and disk performance.

The Acer Aspire 5, equipped with an Intel Core i5 processor, demonstrated adequate performance for basic to moderate computing tasks. However, it struggled with more intensive applications due to its lower core count and core speed. The HP Laptop 15s-fq5xxx, with its Intel Core i7 processor and a combination of performance and efficiency cores, showed strong performance in both single-threaded and multi-threaded tasks. It balanced power consumption and performance, making it a versatile choice for professional use.

The ASUS Zenbook S13 OLED, powered by an AMD Ryzen 7 processor, exhibited superior performance in both single-threaded and multi-threaded applications. Its advanced 6nm technology and higher core count provided excellent efficiency and performance density. This laptop handled workloads more efficiently than the other two, making it ideal for computationally intensive tasks.

The implementation of the CPU testing program further confirmed the laptops' performance capabilities. The results showed a consistent increase in CPU time with higher workloads across all three devices. However, the ASUS Zenbook S13 OLED displayed a slightly lower increase in CPU time, indicating better efficiency in managing growing workloads.

In conclusion, while all three laptops exhibited predictable scaling in processing time with increased workloads, the ASUS Zenbook S13 OLED emerged as the most efficient in handling computationally intensive activities. This project underscores the importance of benchmarking in evaluating and comparing the performance of different CPUs, providing valuable insights for consumers and developers in making informed decisions.

5.0 Reflection

EVELYN GOH YUAN QI

I gained invaluable insights into the principles of CPU performance evaluation through this project. Through hands-on experience with tools like CPU-Z, Geekbench, and MiniTool Partition Wizard, I learned to assess various performance metrics and understand their practical implications. The project highlighted the importance of considering multiple factors such as core count and clock speed in performance analysis. Collaborating with my team to set up benchmarks, run tests, and analyze results taught me the value of effective communication and coordination. This experience not only deepened my understanding of computer architecture but also ignited a keen interest in future advancements in CPU technology.

JOANNE CHING YIN XUAN

I learned a lot about evaluating CPU performance and making good use of benchmarking tools from our Computer Organization & Architecture project. Our teamwork enabled us to analyze data and understand how CPUs execute different jobs, despite challenges like setting up benchmarks on different systems. My understanding of computer architecture has increased as a result of this project, and I'm now more driven to learn about future advances in CPU technology to have a major contribution in this field.

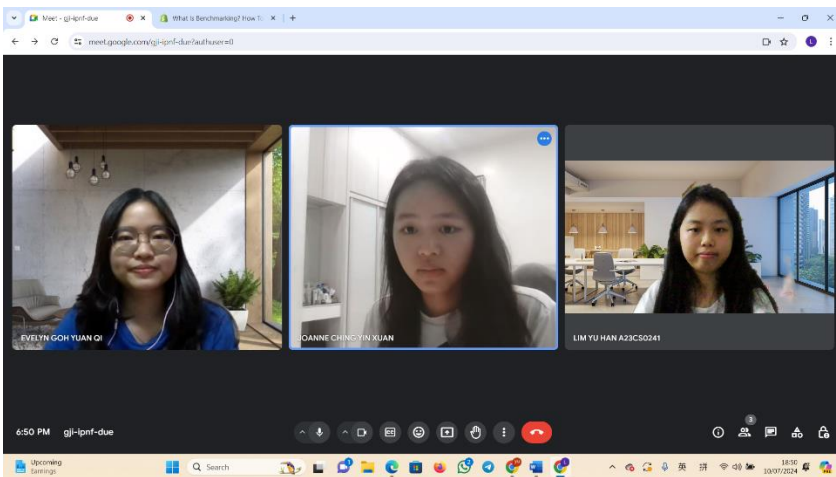
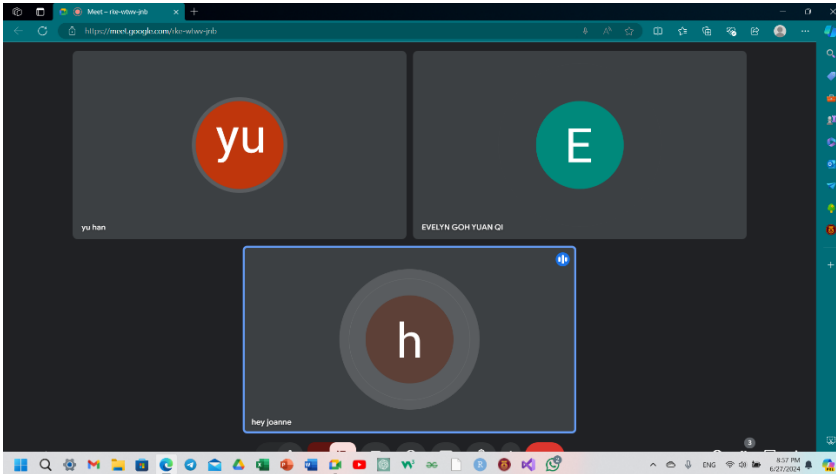
LIM YU HAN

This project taught me how to utilize assembly language to benchmark CPU performance and gain a better understanding of how different systems handle computational tasks because we had to create code in a low-level programming language to measure time before and after loop execution. Furthermore, by comparing the results from multiple laptops, we can establish which systems are more efficient and appropriate for intense computational tasks. This information is useful for consumers like us who want to acquire gear that meets specific standards. I've also honed my problem-solving skills as a result of the challenges encountered in this project, such as data organization, dealing with large numbers, and debugging code. I now fully understand the complexity of CPU efficiency and the factors influencing processing time.

6.0 References

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7.0 Appendix



➔ Forwarded

Intro(problem)-jo
Intro(aim)-yh
Intro(obj)-eve
Intro(scope)-jo
Intro benchmarking(1,2)-yh
Intro benchmarking(3,4)-eve
CPU-Z-jo
Geekbench-yh
Window Task manager-eve
Minitools -jo
Comparison acer -yuhan
Comparison hp-jo
Comparison asus -eve
Discussion cpuz-jo
Discussion geekbench-yh
Discussion task-Eve
Discussion minitools-Jo
Part2 what about+code-yuhan
Result-Eve
Conclusion 1-Jo
Conclusion 2-eve

Reflection own part

17:17 ✓